

Bio503 subjective current Paper solution for final term

Questions

1. What is hydrogen bonding?

Ans:

Hydrogen bonding is a special type of dipole-dipole attraction between molecules. It results from the attractive force between a hydrogen atom covalently bonded to a very electronegative atom such as a N, O, or F atom and another very electronegative atom. Hydrogen bonds are generally stronger than ordinary dipole-dipole and dispersion forces, but weaker than true covalent and ionic bonds. Without hydrogen bonding, earth would have no water cycle, no ocean, no floating ice, no life as we know it.

2. What is NMR?

Ans:

Nuclear magnetic resonance spectroscopy, most commonly known as NMR spectroscopy, is a research technique that exploits the magnetic properties of certain atomic nuclei.

- This type of spectroscopy determines the physical and chemical properties of atoms or the molecules in which they are contained. It relies on the phenomenon of nuclear magnetic resonance and can provide detailed information about the structure, dynamics, reaction state, and chemical environment of molecules.

3. What is millisievert?

The millisievert and milligray as measures of radiation dose and exposure. In the SI system, a millisievert (mSv) is defined as "the average accumulated background radiation dose to an individual for 1 year, exclusive of radon, in the United States.

4. Name the voltage gated Channels

Voltage-gated ion-channels are usually ion-specific, and channels specific to sodium (Na^+), potassium (K^+), calcium (Ca^{2+}), and chloride (Cl^-) ions have been identified. And proton ion channel also.

5. What three spectroscopy techniques

Atomic Emission spectroscopy

It uses the range of electromagnetic spectra in which a substance radiates (emits).

- The substance first must absorb energy. This energy can be from a variety of sources, which determines the name of the subsequent emission, like luminescence.
- Molecular luminescence techniques include spectrofluorimetry

Atomic Absorption spectroscopy

It uses the range of the electromagnetic spectra in which a substance absorbs.

- This includes atomic absorption spectroscopy and various molecular techniques, such as infra-red spectroscopy in that region and Nuclear Magnetic resonance spectroscopy in the radio region

Scattering spectroscopy

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It uses the range of electromagnetic spectra in which a substance radiates (emits).

- The substance first must absorb energy. This energy can be from a variety of sources, which determines the name of the subsequent emission, like luminescence.
- Molecular luminescence techniques include spectrofluorimetry

6. How osmosis occurs in plants.

Root cells of plants absorb water by osmosis. Many fine root hairs increase the surface area the plants roots allowing them to draw in more water from the surrounding soil. The pressure of water in plant cell gives structure to the plants.

7. Biophysical society

1957: Biophysical Society was founded

Aims: to boost the development and distribution of knowledge in biophysics through many activities including meetings, publications, community outreach, and career placement

Members: >9,000, work in academia, industry, and government agencies worldwide

Annual Meeting is held each year in February or March. With over 7,000 attendees

The Society publishes the Biophysical journal twice monthly

8. What is the direct effect of ionizing radiations

Ans: The direct effect of ionizing radiation and the radiant energy is immediately absorbed inside the cellular nucleus.

9. Use of biophysics in future

Ans:

10. Explain types of HPLC

Normal Phase HPLC:

- This method separates analytes on the basis of polarity.
- NP-HPLC uses polar stationary phase and non-polar mobile phase. Therefore, the stationary phase is usually silica and typical mobile phases are hexane, methylene chloride, chloroform, diethyl ether, and mixtures of these.
- Polar samples are thus retained on the polar surface of the column packing longer than less polar materials.

Reverse Phase HPLC:

- The stationary phase is nonpolar (hydrophobic) in nature, while the mobile phase is a polar liquid, such as mixtures of water and methanol or acetonitrile.
- It works on the principle of hydrophobic interactions hence the more nonpolar the material is, the longer it will be retained.

Size-exclusion HPLC:

- The column is filled with material having precisely controlled pore sizes, and the particles are separated according to their molecular size.

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- Larger molecules are rapidly washed through the column; smaller molecules penetrate inside the porous of the packing particles and elute later.

Ion-Exchange HPLC:

- The stationary phase has an ionically charged surface of opposite charge to the sample ions.
- This technique is used almost exclusively with ionic or ionizable samples.
- The stronger the charge on the sample, the stronger it will be attracted to the ionic surface and thus, the longer it will take to elute.
- The mobile phase is an aqueous buffer, where both pH and ionic strength are used to control elution time.

11.Explain third law of thermodynamics .

Ans: The third law of thermodynamics was developed by the German chemist Walther Nernst during the years 1906–12.

□ The third law of thermodynamics is often referred to as Nernst's theorem or Nernst's postulate.

Nernst Heat Theorem (a consequence of the Third Law) is: It is impossible for any process, no matter how idealized, to reduce the entropy of a system to its absolute-zero value in a finite number of operations.

An alternative version of the third law of thermodynamics as stated by Gilbert N. Lewis and Merle Randall in 1923:

“If the entropy of each element in some (perfect) crystalline state be taken as zero at the absolute zero of temperature, every substance has a finite positive entropy; but at the absolute zero of temperature the entropy may become zero, and does so become in the case of perfect crystalline substances.”

The third law of thermodynamics states that the entropy of a system approaches a constant value as the temperature approaches absolute zero.

- The entropy of a system at absolute zero is typically zero, and in all cases is determined only by the number of different ground states it has.
- Specifically, the entropy of a pure crystalline substance (perfect order) at absolute zero temperature is zero.
- This statement holds true if the perfect crystal has only one state with minimum energy.

It says that when we are considering a totally perfect (100% pure) crystalline structure, at absolute zero (0 Kelvin), it will have no entropy (S).

- Note that if the structure in question were not totally crystalline, then although it would only have an extremely small disorder (entropy) in space, we could not precisely say it had no entropy.
- The entropy of a pure perfect crystal is zero (0) at zero Kelvin (0° K).
- Simply putting, the third law of thermodynamics states that if all the thermal motion of molecules (kinetic energy) could be removed, a state called absolute

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zero would occur.

- The Universe will attain absolute zero when all energy and matter is randomly distributed across space.
- The current temperature of empty space in the Universe is about 2.7 Kelvins.

12. Write a note on HPLC instrumentation

Ans: Instrumentation of HPLC

- **Solvent Reservoir** : Mobile phase contents are contained in a glass reservoir. The mobile phase, or solvent, in HPLC is usually a mixture of polar and non-polar liquid components whose respective concentrations are varied depending on the composition of the sample.
- **Pump** : A pump extracts the mobile phase from the solvent reservoir and forces it through the system's column and detector.
- Depending on a number of factors including column dimensions, particle size of the stationary phase, the flow rate and composition of the mobile phase, operating pressures of up to 42000 kPa (about 6000 psi) can be generated.
- **Sample Injector** : The injector can be a single injection or an automated injection system. An injector for an HPLC system should provide injection of the liquid sample within the range of 0.1-100 mL of volume with high reproducibility and under high pressure (up to 4000 psi).
- **Columns** : Columns are usually made of polished stainless steel, are between 50 and 300 mm long and have an internal diameter of between 2 and 5 mm. They are commonly filled with a stationary phase with a particle size of 3–10 μm .
- Columns with internal diameters of less than 2 mm are often referred to as microbore columns. Ideally the temperature of the mobile phase and the column should be kept constant during an analysis.
- **Detector** : The HPLC detector, located at the end of the column detect the analytes as they elute from the chromatographic column. Commonly used detectors are UV spectroscopy, fluorescence, mass-spectrometric and electrochemical detectors.
- **Data Collection Devices** : Signals from the detector may be collected on chart recorders or electronic integrators that vary in complexity and in their ability to process, store and reprocess chromatographic data. The computer integrates the response of the detector to each component and places it into a chromatograph that is easy to read and interpret.

13. Types of protein.

There are four types of proteins:

- **Fibrous**
- **Globular**
- **Simple**

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- **Conjugated**

Fibrous proteins:

In this type the Polypeptides arranged in long strands or sheets. Water insoluble (lots of hydrophobic AA's)

- Strong but flexible
- Structural (keratin, collagen)

Globular proteins:

Polypeptide chains folded into spherical or globular form

- Water soluble
- Contain several

Types of secondary structure

Diverse functions (enzymes, regulatory proteins)

Simple proteins:

- Composed only of amino acid residues

Conjugated proteins:

Contain prosthetic groups (metal ions, cofactors, lipids, carbohydrates)

Example:

Hemoglobin – Heme

15. Shortly explain electrodialysis (3)

Electro dialysis (ED) is used to transport salt ions from one solution through ion-exchange membranes to another solution under the influence of an applied electric potential difference. This is done in a configuration called an electro dialysis cell. The cell consists of a feed (dilute) compartment and a concentrate (brine) compartment formed by an anion exchange membrane and a cation exchange membrane placed between two electrodes. In almost all practical electro-dialysis processes, multiple electrodialysis cells are arranged into a configuration called an electrodialysis stack, with alternating anion and cation exchange membranes forming the multiple electro-dialysis cells.

16. Name two polar compounds?

NaCl, H₂O, H-Cl, H-I

17. Define motion?

In physics, motion is a change in position of an object over time. • Motion is described in terms of displacement, distance, velocity, acceleration, time, and speed. • Motion of a body is observed by attaching a frame of reference to an observer and measuring the change in position of the body relative to that frame.

18. Define energy coupling?

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Energy Coupling:

Living organisms have the ability to couple exergonic and endergonic reactions. Energy released by exergonic reactions is captured and used to make ATP from ADP and Pi

- ATP can be broken back down to ADP and Pi, releasing energy to power the cell's endergonic reactions.
- ATP (adenosine triphosphate) is the cell's energy shuttle.
- ATP is composed of ribose (a sugar), adenine (a nitrogenous base), and three phosphate groups.

19. Explain third law of motion?

Whenever one body exerts a force F onto a second body, (in some cases, which is standing still) the second body exerts the force $-F$ on the first body. F and $-F$ are equal in magnitude and opposite in sense. So, the body which exerts F will go backwards.

20. Explain intermolecular Hydrogen bonding?

Hydrogen bonding :

- Hydrogen bonding is a special type of dipole-dipole attraction between molecules.
- It results from the attractive force between a hydrogen atom covalently bonded to a very electronegative atom such as a N, O, or F atom and another very electronegative atom.
- Hydrogen bonds are generally stronger than ordinary dipole-dipole and dispersion forces, but weaker than true covalent and ionic bonds.
- Without hydrogen bonding, earth would have no water cycle, no ocean, no floating ice, no life as we know it.

21. Application of biophysics?

- Application to the study of biological molecules.
- Develop quantitative and physical understanding of molecular and cellular aspects of biology and the underlying logic of living systems.
- Learn laboratory skills and technique for analysis of biomolecules, with emphasis on structure-function relationships.
- Investigation of radiation effects on living matter. Physicists were introduced to biologists and
- Biological problems. Biophysics evolved as a new scientific field.

22. Define echolocation?

Echolocation is the use of sound waves and echoes to determine where objects are in space.

- Echolocation, also called bio sonar, is the biological sonar used by several kinds of animals.

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- Echo-locating animals emit calls out to the environment and listen to the echoes of those calls that return from various objects near them.
- They use these echoes to locate and identify the objects.
- Echolocation is used for navigation and for foraging (or hunting) in various environments.

23. Name the two voltage gated ionchannels?

Voltage-gated ion-channels are usually ion-specific, and channels specific to sodium (Na^+) channel, potassium (K^+) channel, calcium (Ca^{2+}) channel, and chloride (Cl^-) channel ions have been identified.

23. What is Channelopathies?

Channelopathies is a disease caused due to the mutations of ion channel. Channelopathies are diseases caused by disturbed function of ion channel subunits or the proteins that regulate them.

- These diseases may be either congenital (often resulting from a mutation or mutations in the encoding genes) or acquired (often resulting from autoimmune attack on an ion channel).

23. What is radiation protection?

Defined by the International Atomic Energy Agency (IAEA) as "The protection of people from harmful effects of exposure to ionizing radiation, and the means for achieving this".

The IAEA also states

"The accepted understanding of the term radiation protection is restricted to protection of people. Suggestions to extend the definition to include the protection of non-human species or the protection of the environment are controversial". Exposure can be from a radiation source external to the human body or due to the bodily intake of a radioactive material.

Ionizing radiation is widely used in industry and medicine, and can present a significant health hazard by causing microscopic damage to living tissue. This can result in skin burns and radiation sickness at high exposures, known as "tissue" or "deterministic" effects (conventionally indicated by the gray), and statistically elevated risks of cancer at low exposures, known as "stochastic effects" (conventionally measured by the sievert).

24. What is cation and anion exchanger?

Anion exchange:

In a buffer with a pH greater than the pI of the protein of interest, the protein will carry a net negative charge; therefore, a positively charged anion exchange resin is chosen to capture this protein.

• Cation exchange:

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In a buffer with a pH lower than the pI of the protein of interest, the protein will carry a positive net charge; thus a negatively-charged cation exchange resin is chosen. When an ion exchange chromatography column is loaded with a sample at a particular pH, all proteins.

25. What is HAZMAT?

Materials (HAZMAT)

Response **team** is comprised of **hazardous material** experts who specialize in detecting, containing, and removing any release or potential release of hazardous substances in order to control or stabilize an incident.

What is Enthalpy of mixing?

Enthalpy of mixing, defined as the enthalpy change upon mixing of two (non-reacting) chemical substances. When a substance or compound is combined with any other substance or compound the enthalpy of mixing is the consequence of the new interactions between the two substances or compounds. This enthalpy if released exothermically can in an extreme case cause an explosion.

26. Five types of process?

Quasi-equilibrium process:

Quasi means almost and static means the thermodynamic properties static with time. In thermodynamics, a quasi-static process is a thermodynamic process that happens infinitely slowly. An infinitely slow process such that the system undergoes changes in its variables (P, V, T) so slowly that it continues to be in its thermal, mechanical and chemical equilibrium with its surroundings all the time and at all points.

Non-equilibrium process:

The system is not in equilibrium during the process.

States during the process are undefined. Process carried out normally, without any attempt at limiting irreversibility by slowing it down. All natural processes occurring in the universe are irreversible and non-quasi-static.

27. Write parts of HPLC?

- **Solvent Reservoir**
- **Pump**
- **Sample Injector**
- **Columns**
- **Detector**
- **Data Collection Devices**

28. What are Biophysical Techniques?

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- The characterization of molecular structure, the measurement of molecular properties, and the observation of molecular behavior presents an enormous challenge for biological scientists. A wide range of biophysical techniques have been developed to study molecules in crystals, in solution, in cells, and in organisms.
- Biophysical techniques provide information about the electronic structure, size, shape, dynamics, polarity, and modes of interaction of biological molecules.
- Some of the most exciting techniques provide images of cells, subcellular structures, and even individual molecules.
- Directly observe the biological behavior and physical properties of single protein or DNA molecules within a living cell
- Determine how the behavior of the single molecule influences the biological function of the organism.

29. What is ball bearing?

Ball bearing is a device which consists of a ring of small metal balls. The small metal balls of a ball bearing can roll freely. They are designed to make the moving parts of a machine to roll over each other rather than slide. The ball bearing is introduced between the two surfaces which have to rotate over each other. **Friction can be reduced by using ball bearing between the moving parts of machine.**

30. How diffusion is done in lungs?

Diffusion in Lungs:

Diffusion of O₂ into the alveoli of the lungs. Diffusion is important to organisms because it is the process by which useful molecules enter the body cells and waste products are removed. Oxygen moves from high concentration (in the air sac) to a lower concentration (in the blood). Carbon dioxide moves from high concentration (in the blood) to a lower concentration (in the air sac).

31. What is direct and indirect effect ionizing radiation?

The **direct effect** of ionizing radiation and the radiant energy is immediately absorbed inside the cellular nucleus.

The **indirect effect** induces radiolysis of water, which produces very aggressive free radicals responsible for radiation damage of important molecules. The absorption of radiation in biological environment depends on high water content.

32. What is Random Walk?

A random walk is a mathematical object, known as random process, that describes a path that consists of a succession of random steps on some mathematical space such as the integers. An elementary example of a random walk is the random walk

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on the integer number line Z , which starts at 0 and at each step moves +1 or -1 with equal probability. A random walk is the process by which randomly moving objects wander away from where they started.

33. Write Difference between passive and active **electrolocation**?

Active electrolocation	Passive electrolocation
Active electrolocation Conductive objects concentrate the field and resistive objects spread the field In active electrolocation, the animal senses its surrounding environment by generating electric fields and detecting distortions in these fields using electroreceptor organs. • This electric field is generated by means of a specialized electric organ consisting of modified muscle or nerves. • This field may be modulated so that its frequency and wave form are unique to the species and sometimes, the individual.	• In passive electrolocation, the animal senses the weak bioelectric fields generated by other animals and uses it to locate them. • These electric fields are generated by all animals due to the activity of their nerves and muscles. A second source of electric fields in fish is the ion pumps associated with osmoregulation at the gill membrane. This field is modulated by the opening and closing of the mouth and gill slits. • Many fish that prey on electrogenic fish use the discharges of their prey to detect them. This has driven the prey to evolve more complex or higher frequency signals that are harder to detect. • Passive electroreception is carried out solely by ampullary electroreceptors in fish. It is tuned to low frequency signals (below one up to tens of Hertz) • Fish use passive electroreception to supplement